

AdventureMultiplayer - Power-Up Reference

1. Self-Targeted Power-Ups

SpeedBoost

Effect:	topSpeed x1.4 acceleration & airAcceleration x1.2
Duration:	5 seconds (SpeedBoostDur)
Applied to:	Owner client only via BeginSpeedBoostClientRpc
Stacking:	Stacks with Rocket - each has its own cancellation token

Rocket

Effect:	topSpeed x2.0 acceleration & airAcceleration x2.0
Duration:	3 seconds (RocketDur)
Applied to:	Owner client only via BeginRocketClientRpc
Stacking:	Stacks with SpeedBoost - each has its own cancellation token
Note:	Per-player SO clones created in Start() - scaling never leaks to other players

Shield

Effect:	Absorbs the next ONE adverse effect of any type
Max duration:	30 seconds (ShieldMaxDur) - auto-expires if no hit is taken
Consumed on:	First hit absorbed; shield is gone after a single block
Blocked types:	Damage, StunBolt, Swap, Freeze (Banana not checked, DecoyBox bypasses)
Replicated:	NetworkedHealth.ShieldActive (NetworkVariable<bool>) - visible to all clients

Invincibility

Effect:	Blocks all damage, stun, swap, freeze, and banana slip
Duration:	2 seconds (InvincDur)
Consumption:	Not consumed per hit - runs a flat time window
Replicated:	NetworkedHealth.InvincibleActive (NetworkVariable<bool>) - visible to all clients

2. Targeted Power-Ups (Offensive)

StunBolt

Target:	Player directly ahead in race position
If in 1st:	Targets the player directly behind instead
Effect:	StunEffect - zeros lateral & vertical velocity
Duration:	3 seconds (StunBoltDur)
Countered by:	Shield (consumed) or Invincibility (not consumed)

Swap

Target:	Player directly ahead only; no effect if already in 1st
Effect:	Teleports caster and target to each other's world positions
Race scoring:	Swaps checkpoint indices in RaceManager - scoring reflects swap immediately
Countered by:	Shield on target (consumed) or Invincibility on target (not consumed)

Freeze

Target:	ALL players currently behind the caster
Effect:	StunEffect on each target - zeros velocity
Duration:	1 second (FreezeDur)
Countered by:	Shield per target (each shield consumed individually) or Invincibility

3. World-Trap Power-Ups

Banana Peel

Spawns:	1.2 units behind the caster at ground level
Lifetime:	15 seconds (BananaPeel.lifetime)
Effect:	SlipEffect - forward-only movement, no turning, no jumping
Slip duration:	2.5 seconds (BananaSlipDur)
Trigger:	One-shot - despawns after first contact
Owner immune:	Yes - placer is always immune to their own peel
Countered by:	Invincibility on the player who walks over it

Decoy Box

Spawns:	3 units in front of caster, 0.5 m above ground
Lifetime:	20 seconds (DecoyBoxPickup.lifetime)
Effect:	StunEffect - full movement freeze
Stun duration:	2 seconds (DecoyStunDur)
Trigger:	One-shot - despawns after first contact
Owner immune:	Yes - placer is always immune
Bypasses:	Both Shield and Invincibility - intentional design, cannot be blocked

4. Counter Table

Incoming Effect	Shield	Invincibility
Health damage (obstacle / hazard)	Absorbs - consumed	Blocks - not consumed
StunBolt	Absorbs - consumed	Blocks - not consumed
Swap	Absorbs - consumed	Blocks - not consumed
Freeze	Absorbs - consumed	Blocks - not consumed
Banana slip	Not checked - passes through	Blocks - not consumed
Decoy Box stun	Bypasses by design	Bypasses by design

5. Timing & Duration Quick Reference

Power-Up	Duration	Constant in Code
SpeedBoost	5 s	SpeedBoostDur
Rocket	3 s	RocketDur
Shield (max)	30 s	ShieldMaxDur
Invincibility	2 s	InvincDur
StunBolt stun	3 s	StunBoltDur
Freeze stun	1 s	FreezeDur
Banana slip	2.5 s	BananaSlipDur
Decoy Box stun	2 s	DecoyStunDur
Banana Peel (trap)	15 s	BananaPeel.lifetime
Decoy Box (trap)	20 s	DecoyBoxPickup.lifetime

6. Key Implementation Files

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File	Responsibility
PlayerPowerUpInventory.cs	3-slot inventory, UseSlotServerRpc, all dispatch logic
PowerUpEnums.cs	PowerUpType enum, PowerUpLevelData struct
StunEffect.cs	Velocity-zeroing stun applied on owner client
SlipEffect.cs	Forward-only movement applied on owner client
BananaPeel.cs	World-trap: one-shot trigger, slip on contact
DecoyBoxPickup.cs	World-trap: one-shot trigger, stun, bypasses defences
NetworkedHealth.cs	SetShield, SetInvincible, ShieldActive / InvincibleActive NetworkVars
NetworkedPowerUpBox.cs	Collectible boxes that grant power-ups on contact
PowerUpSlotHUD.cs	Slot display and tap-to-use buttons
PowerUpUpgradeManager.cs	Level scaling (not yet wired to level constants)